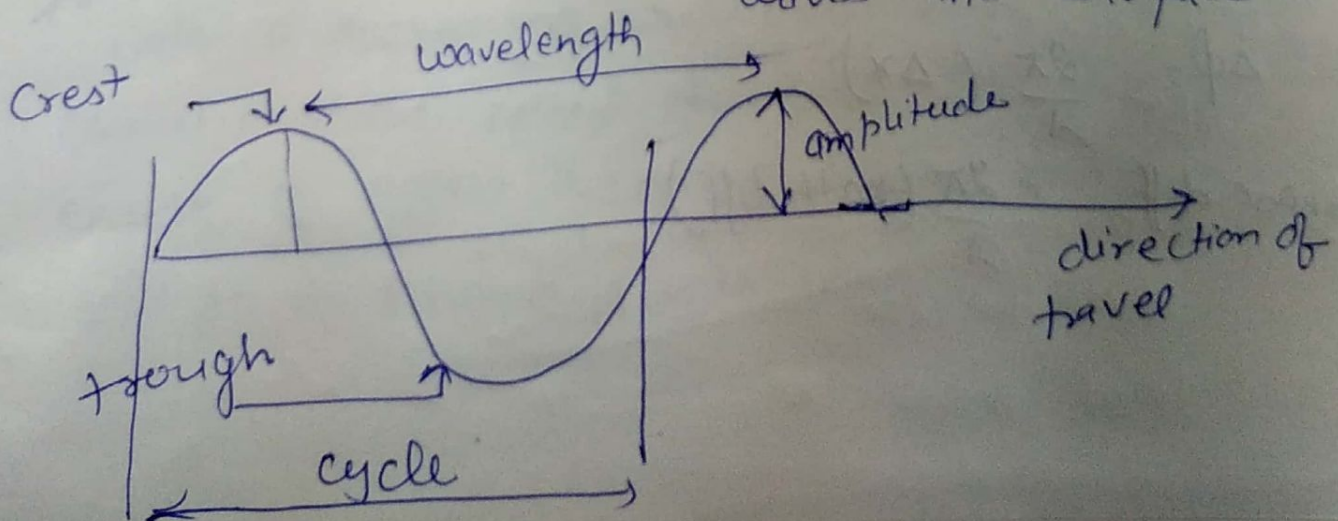


## ★ Waves and Oscillations:

Wave motion: - It is defined as the energy and momentum movement from one point to another in a medium instead of actually transferring the substances between the places. In other words, it is the propagation of disturbances across a medium that is caused by repetitive vibrations of particles around their mean locations. Waves create disturbance as a function of time, and this function is referred to as the wave function. In the case of string, it may be particle displacement and in the case of sound waves, it could be a change in pressure or density.

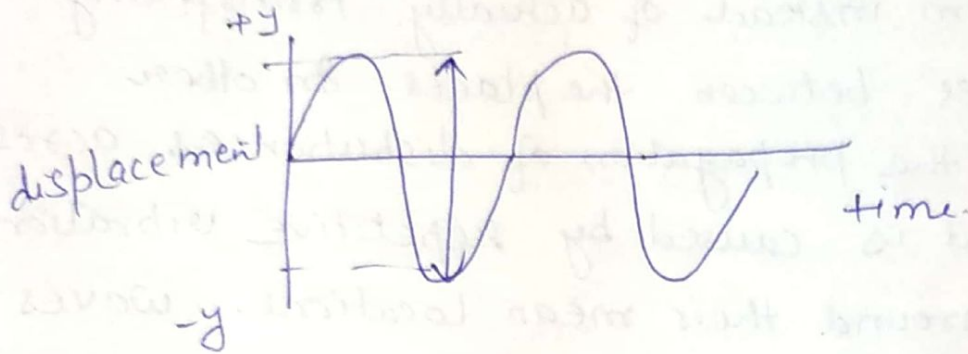
## ★ Types of waves: -

1. Mechanical or Elastic waves → requires medium
  2. Electromagnetic waves (Non-Mechanical waves)
  3. Matter waves
- (The particles present in the medium exert a periodic motion at the mean position during the propagation of wave within medium)
- ↓  
Transverse waves, longitudinal waves and surface waves

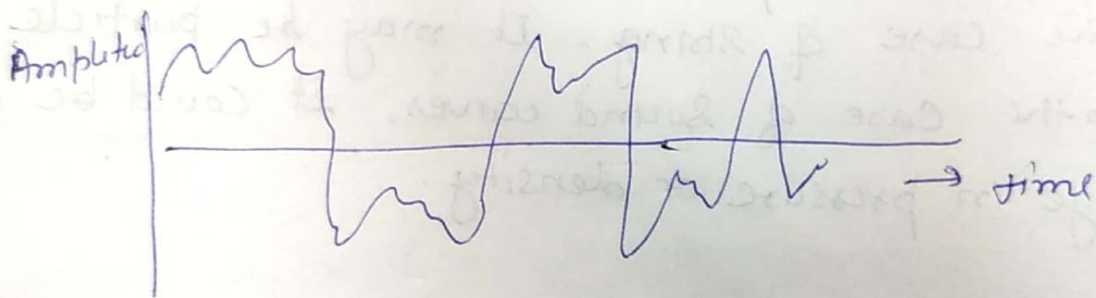




★ Periodic Waves — if the disturbance is continuous and periodic in character. A sinusoidal periodic wave is a periodic wave that varies sinusoidally. When a sinusoidal periodic wave travels through the medium, the particles perform SHM.



★ Non-periodic wave : —



★  $\text{Velocity} = \frac{1}{T} = f$   $\xrightarrow{\text{freq.}}$

2.  $\omega = \text{ang. freq.} = \frac{2\pi}{1} = 2\pi f$

3. Path & phase difference : —  
 $\Delta x$   $\Delta \phi$

$$\Delta \phi = \frac{2\pi}{\lambda} (\Delta x)$$

$$\text{phase diff} = \frac{2\pi}{\lambda} (\text{path diff})$$



★ Simple Harmonic Motion :- It is a special type of periodic motion, in which a particle moves to and fro repeatedly about a mean position and magnitude of force acting on particle at any instant is directly proportional to the displacement of particle from mean position at that instant

$$F = -Ky$$

$K$  is force constant, -ve sign shows that the restoring force ( $F$ ) is always directed towards the mean position.

→ The displacement of particle executing S.H.M. at an instant is defined as the distance of particle from the mean position.

$$y = a \sin \omega t \quad \text{or} \quad y = a \cos \omega t.$$

$$\frac{dy}{dt} = \text{velocity} = a\omega \cos \omega t = a\omega \sqrt{1 - \sin^2 \omega t} \\ = \omega \sqrt{a^2 - y^2}$$

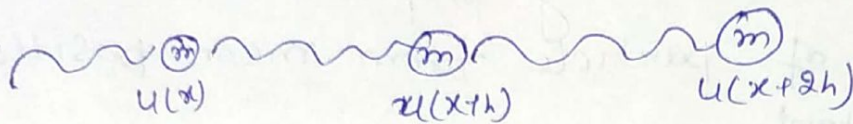
★ When a Simple Harmonic system oscillates with a decreasing amplitude with time, its oscillations are called damped oscillations. The energy of system executing damped oscillation will go on decreasing with time.



Wave equation :-

Derived from Hooke's law

$$F = -kx$$



array of mass  $m$  is interconnected with massless spring of length  $h$  with spring const.  $k$

$u(x)$  measures the distance from eq<sup>m</sup> of mass situated at  $x$ . The forces exerted on mass  $m$  at the location  $x+h$  are

$$F = m \cdot a(t) = m \cdot \frac{\partial^2 u(x+h, t)}{\partial t^2} \quad \text{--- (1)}$$

$$F_{\text{Hooke}} = F_{x+2h} + F_x = k[u(x+2h, t) + u(x, t)] \\ = k[u(x+2h, t) - u(x+h, t)] + k[u(x, t) - u(x+h, t)] \quad \text{--- (2)}$$

The equation of motion for the weight at the location  $x+h$  is given by equating these two forces.

$$m \frac{\partial^2 u(x+h, t)}{\partial t^2} = k[u(x+2h, t) - u(x+h, t) + u(x, t) - u(x+h, t)]$$

If the array consists  $N$  weights spaced evenly over the length  $L = N \cdot h$

and total mass  $M = N \cdot m$  &  $k = K/N$

So above equation becomes

$$\frac{\partial^2 u(x,t)}{\partial t^2} = \frac{KL^2}{M} \frac{[u(x+2h,t) - 2u(x+h,t) + u(x,t)]}{h^2}$$

Taking the limit  $N \rightarrow \infty$ ,  
 $h \rightarrow 0$  (and assuming smoothness)

$$\frac{\partial^2 u(x,t)}{\partial t^2} = \frac{KL^2}{M} \frac{\partial^2 u(x,t)}{\partial x^2}$$

$\frac{KL^2}{M}$  is the square of the propagation speed  
in this particular case. Taking  $c^2 = \frac{KL^2}{M}$  we  
have the one dimensional wave equation as

$$\boxed{\frac{\partial^2 u(x,t)}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 u(x,t)}{\partial t^2}}$$